Ask Me Nothing: Learning Need-to-Know Clarification for Financial Agents with Multi-Turn RL

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Anchored in: L3 (Dialogue as a POMDP), L10 (Post-Training), L12 (Evaluation and User Simulation), L15 (Reasoning and Test-Time Compute), L16 (Agents, Tool Use, and Agentic RL).

1 Problem and Why It Is Open

In financial dialogue every clarifying question is also a request for client data. A brokerage agent that asks by habit wastes the client's time, loses money while prices move, and collects sensitive information it never needed; one that never asks guesses at goals it cannot see. Regulation pulls in both directions: FINRA Rule 2111 requires a broker to use reasonable diligence to ascertain a customer's investment profile (Financial Industry Regulatory Authority, 2012), while the SEC's 2024 amendments to Regulation S-P raise the cost of holding customer information that is later exposed (Financial Industry Regulatory Authority, 2024). The agent should therefore ask *exactly* what it needs to know: look up or derive what it can, ask what the rules require or only the client knows, and pass nothing on.

Closest prior work. Work on clarification mostly targets *under*-asking. UserBench finds that frontier agents uncover fewer than 30% of user preferences through interaction (Qian et al., 2025); Edwards and Schuster (2026) study when coding agents should ask; CollabLLM trains models with a multiturn-aware reward to ask more (Wu et al., 2025), using a GPT-4o-mini user simulator in the training loop, no tools, no policy constraints, and mostly proxy outcomes. τ-bench has the structure we need (hidden user goal, policy document, end-state check) but is an evaluation benchmark, not a training environment (Yao et al., 2025). On privacy, AirGapAgent minimizes data with a separate prompted minimizer and no training (Bagdasarian et al., 2024); AgentDAM measures data minimization in web agents, where a privacy prompt costs utility and training is left to future work (Zharmagambetov et al., 2025); AgentRaft detects data over-exposure across tool calls (Lin et al., 2026); PrivacyLens tests privacy-norm awareness of agent actions (Shao et al., 2024a); AgentDojo measures prompt injection (Debenedetti et al., 2024).

The gap. No prior work *trains* the ask, look-up, think, or act decision with RL in a policy-constrained, tool-using dialogue whose outcome is checked exactly, nor treats each question as a priced data request so that clarification and data minimization are learned as one behavior.

2 Research Questions

- **RQ1.** Can multi-turn GRPO (Shao et al., 2024b) teach a small model when to ask, look up, think, or act? *H1:* it beats the same model prompted and SFT-distilled from a frontier model on success, pass^k, and dollars lost. *Refuted if* it does not beat distillation, or gains come with more policy violations.
- **RQ2.** Does pricing sensitive data teach data minimization? *H2:* fewer requests for unneeded sensitive fields and fewer leaks, with no drop in success. *Refuted if* over-collection falls only because mandatory profile items are skipped or more tasks fail.
- **RQ3.** What did training change? *H3:* gains decompose into fewer redundant questions, retained mandatory ones, and reasoning spent on the turns that need it, not one shortcut such as never asking.
- **RQ4.** Does the policy hold up off distribution? *H4:* it transfers from the scripted training client to LLM-simulated clients and resists injected requests for other clients' data.

3 Method and System

BrokerDesk. We build a simulated brokerage desk. The agent calls tools to read positions, pro-

files, and quotes, run risk checks, place orders, and ask the client, plus third-party tools (counterparty messages, research requests) that create outbound leak opportunities. A policy document sets position limits, a restricted list, order-size and margin rules; prices move between turns. Tasks (rebalance, hedge, raise cash, tax-aware trim) are generated with every needed fact labeled by where it lives: *derivable*, *on file*, *mandatory* to ask, *client-only*, or *irrelevant*. Every question can thus be scored as needed or unneeded. Each client field has a sensitivity tier (low, sensitive, restricted), and every synthetic value carries a unique canary string, so a leak is an exact string match in the transcript or tool arguments, with no LLM judge.

Reward. For an episode with n_q questions, unneeded fields U, n_v policy violations and n_ℓ leaks,

$$R = \mathbf{1}[\text{goal met}] - \lambda_q n_q \\ - \textstyle\sum_{f \in U} \lambda_{\text{tier}(f)} - \lambda_v n_v - \lambda_\ell n_\ell.$$

Training. GRPO with LoRA on Qwen3-1.7B (Qwen3-0.6B for smoke tests, Qwen3-4B if compute allows), whose switchable thinking mode supports the thinking ablation. Training uses a fast, deterministic scripted client that answers truthfully from the hidden goal, with a curriculum from fully specified to underspecified tasks.

4 Data, Baselines, and Evaluation

Data. All clients, portfolios, and tasks are synthetic and generated with fixed seeds: about 300 training and 100 held-out tasks. No human subjects and no real customer data.

Baselines (none designed to lose): (i) the same model with a privacy-aware prompt; (ii) the same model SFT-distilled from filtered frontier-model trajectories; (iii) a prompted frontier model as a ceiling; (iv) an AirGapAgent-style minimizer in front of (i).

Metrics. Success and pass^k (Yao et al., 2025); dollars lost; turns; needed vs. unneeded questions; missed mandatory items; over-collection by tier; leak rate; injection success rate.

Multi-turn evaluation. Every score is computed over whole episodes from the final portfolio state and the full transcript. Held-out tasks are run against the scripted client, LLM-simulated clients, and noisy variants (terse, verbose, once mistaken, pushy), with injected data requests in tool outputs. We report 3 seeds with 95% confidence intervals, broken down by client type and information class. **Ablations:** privacy cost off; question cost off; thinking off, on, or per turn; look-up tools removed.

5 Compute Plan

Training needs one A100/H100-class GPU on PACE-ICE, where the team already runs GPU jobs for course assignments. Our estimate: about 5 to 10 GPU-hours per GRPO run at 1.7B, so 3 seeds of the main run plus 3 ablations is roughly 50 to 100 GPU-hours. Prompted baselines, distillation data, and LLM clients use API calls with a capped budget we report in the paper.

6 Risks and Fallback

No learning signal (checked on a 0.6B run by Nov 15): easier curriculum and denser shaping. **Reward hacking** (never asking or never acting): detected by ask rates on fully specified tasks and penalized. **Privacy–success trade-off**: we report the Pareto frontier across λ_{tier} settings as a finding. **Limited GPUs**: drop to 0.6B and fewer ablations. **Fallback that still yields a paper:** BrokerDesk plus an API-only study of how frontier and prompted models price questions and leak data (RQ2 to RQ4 without training), which is a complete contribution if RL underperforms.

7 Division of Labor

Member	Workstream
Greyson	BrokerDesk environment, tools, policy checker
Heeba	Clients, task generator, privacy tiers, leak checks
Michael	GRPO training on PACE-ICE, ablations
Canting	Baselines, evaluation, paper drafting

Everyone reviews results and writes. Minimal BrokerDesk runs end to end by Nov 9; first RL result and baselines by the Nov 19 milestone; main runs and ablations by Dec 2. Code: https://github.com/IncorvaiaM1/ask-me-nothing.

AI Use

Following ACL's policy on generative assistance, we disclose that we used Claude for literature search, brainstorming (including model-suggested ideas that we evaluated and adopted), and drafting help; all authors checked every claim and citation.

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